



HACKSTREET BOYS

FixHub: A Community-Based Maintenance Reporting System

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ABSTRACT



FixHub is a community-based maintenance and residential management system developed to modernize how tenants and administrators handle service requests, complaints, negotiations, and dormitory-related communications. Designed specifically for shared housing environments. The system replaces traditional manual reporting methods with an integrated digital platform that promotes transparency, efficiency, and timely response to tenant concerns.

- TO PROVIDE A CHATBOX FEATURE FOR DIRECT AND IMMEDIATE TENANT-MANAGEMENT COMMUNICATION.
- TO ENABLE OCCUPANCY TRACKING AND MONITORING FOR LANDLORD USE.

- TO CREATE MODULES FOR COMPLAINTS, RULES, AND BLOCKLIST MANAGEMENT TO ENHANCE COMMUNITY GOVERNANCE.
- TO AUTOMATE MAINTENANCE AND SERVICE REQUESTS USING AN INTEGRATED REQUEST BUTTON AND TRACKING SYSTEM.
- TO IMPLEMENT A NEGOTIATION MODULE ALLOWING TENANTS TO MAKE COUNTER-OFFERS AND DISCUSS SERVICE TERMS DIGITALLY.

FIXHUB

METHODOLOGY (DESCRIPTIVE-DEVELOPMENTAL RESEARCH DESIGN)

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Phase 1: Requirement Analysis: Researchers gathered data from CJKP Dormitory tenants and landlords to identify challenges in manual maintenance reporting and communication gaps.

Phase 2: System Design: Utilized a Descriptive-Developmental Research Design and the Input-Process-Output (IPO) model to structure the system's features and workflow.

Phase 3: Prototype Development: Built the system using Python (Django) and SQLite, integrating core modules for requests, negotiations, and real-time chat.

Phase 4: Quantitative Evaluation: Conducted an evaluation with 20 tenants and IT experts using the Technology Acceptance Model (TAM) and ISO 25010 standards to assess usability.

Phase 5: Implementation & Deployment: Deployed the system for pilot testing at the dormitory to validate its effectiveness in improving operational transparency and efficiency.

RESULTS

FixHub achieved Very High Acceptance (Overall TAM Mean: 3.82).
Exceptional Perceived Usefulness (3.83) in streamlining dormitory operations and maintenance reporting.
Tenants showed strong willingness to continue using the system (Highest Mean: 3.85).
Features effectively automated maintenance requests, negotiations, and emergency alerts.
System demonstrated technical robustness with a Very High ISO 25010 rating (3.39).

CONCLUSION

Successfully addresses the communication gaps and inefficiencies of manual reporting at CJKP Dormitory.
Validated as a technically viable and user-friendly platform specifically tailored to shared residential needs.
High user acceptance scores (3.82 Mean) confirm that tenants are ready and highly willing to adopt the system for daily use.
Empowers administrators by providing real-time, transparent data for maintenance tracking and rule enforcement.
Transforms traditional verbal reporting into a streamlined, digital, and data-driven community management system.

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